Muhammad Hassan

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Experience

HR Mobiles
System Engineer **FEB 2024- JUN 2025**
Designed, deployed, and maintained DMR communication, switching, and routing systems, performing network troubleshooting, site assessments, and performance optimization to ensure reliable connectivity.

RADAR LAB (SINES, NUST).
Research Assistant **SEP 2022 – JUN 2023**
Developed a posture recognition and fall detection system using mm-Wave radar and neural networks, applying data preprocessing, feature extraction, model training, and evaluation for human activity recognition.

Education

DEBRECENI EGYETEM **SEP 2025 - Current**
Master's of Science in Data Science
Courses: Natural Language Processing, Data Minning, Optimization Algorithms, Machine Learning
Thesis: Traffic Simulation and Analysis for Smart Cities: Leveraging Graph-Based Stochastic Simulation and Geospatial Analytics For Real-Time Traffic Analysis and Congestion Prediction in Smart Cities

National University of Sciences and Technology (NUST) **SEP 2019- JUN 2023**
Bachelor's in Electrical Engineering
Courses: Calculus, Signals & Systems, Microwave Devices, CCNA
Final Year Project: Posture Recognition and Fall Detection: Fusing mm-Wave Radar Technology with Neural Networks to differentiate between different postures

Skills

- **Scientific Computing**: Python, R, C/C++, MATLAB, SQL, Data Processing
- **AI, Deep Learning**: PyTorch, TensorFlow, Keras, Scikit-Learn
- **Foundation Models, GenAI**: GPT, Multimodal Llama, RAG, and AI Assistants

Projects:

NLP Multi-Agent Synthesizer & Reviewer | Llama 3.1 (8B) **Spring 2026**
Developed a multi-agent NLP pipeline with specialized agents for text analysis, cross-agent validation, and iterative refinement, benchmarking performance against a single-prompt baseline.

Data Mining for Air Quality Assessment and AQI Prediction Using RapidMiner **Fall 2025**

Applied RapidMiner for air-quality preprocessing, feature selection, clustering, and classification to identify pollution patterns and predict AQI categories from pollutant and meteorological data.

Honors and Rewards:

- Stipendium Hungaricum Scholarship
- Prime Minister’s Ehsaas Undergraduate Scholarship (fully funded award for complete BS degree)
- Awarded $1000 NGIRI Ignite funding for Final Year Project